High Voltage Techniques Presentation

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Determination of Voltage Distribution and Flashover Voltage for Insulator Strings

The purpose of the test

- As we know the string of suspension insulators consists of a number of porcelain discs connected in series through metal links. The porcelain portion of each disc is in between two metal links. Thus, the disc acts like a capacitor. The capacitance is called the **mutual capacitance**. In actual practice, a capacitance also exists between the metal fitting of each disc and the tower. This capacitance is called the shunt capacitance
- Due to the presence of the shunt capacitance, the charging current is not the same through all the discs of the string. The voltage across each disc will be different. The disc nearest to the line conductor will have the maximum voltage.

The purpose of the test

- The insulators, which are used for the suspension of overhead transmission lines, constitute one of the most important parts of the transmission lines as flashover effects in insulators can cause the tripout of a transmission network. Capacitances are developed between each insulator in the insulator string with respect to the high voltage conductor as well as to earth. In addtion, stray capacitances have different values for different positions of the insulators in the string. As a result, the voltage distribution along the insulators is not uniform, with insulators nearer the conductor are more highly stressed.
- Uneven voltage distribution and the failure of insulators may occur hence the neccessity to measure the voltage distribution along the insulator string.

The equipment used in the test

Equipment used in the test:

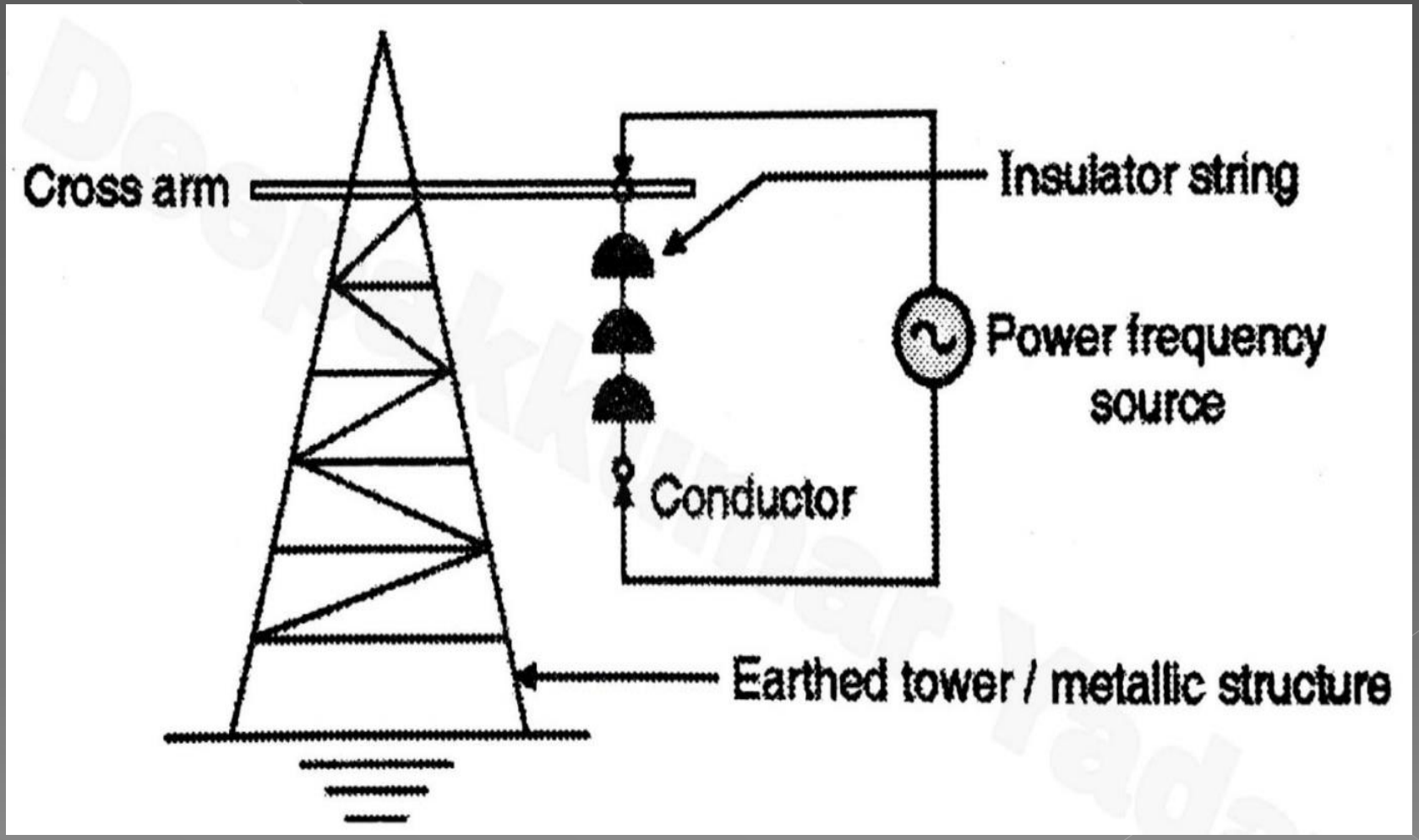
- High Voltage Power Supply: This device generates the high voltage required for the test. It is capable of supplying a range of voltages depending on the specific requirements of the test.
- Insulator Strings: These are the insulating components that are connected in series to form a string. They are designed to withstand high voltages and prevent electrical breakdown.
- Potential Dividers: These devices are used to measure the voltage distribution along the insulator string. They provide voltage taps at specific intervals to measure the voltage at different points along the string.
- Voltage Measurement Devices: These instruments, such as digital voltmeters, are used to measure the voltage at various points on the insulator string and potential dividers.
- Current Measurement Devices: These instruments, such as current transformers, are used to measure the current flowing through the insulator string during the test.

The equipment used in the test

Test Setup:

- Insulator String Arrangement: The insulator strings are arranged in a specific configuration, typically in a vertical or inclined arrangement, depending on the test requirements.
- High Voltage Power Supply Connection: The high voltage power supply is connected to one end of the insulator string.
- Potential Divider Placement: Potential dividers are connected at specific intervals along the insulator string, usually at equal distances, to measure the voltage distribution.
- Voltage and Current Measurements: Voltage and current measurement devices are connected to the potential dividers and insulator string to monitor and record the voltage and current levels during the test.
- Grounding: Proper grounding measures are implemented to ensure safety and proper functioning of the setup.
- Test Chamber or Insulator Support Structure: In some cases, a test chamber or an insulator support structure may be used to provide mechanical support and proper positioning of the insulator strings during the test.

The equipment used in the test



Voltage Distribution and Flashover Voltage

- Voltage distribution refers to the distribution of voltage values at different points in an electrical system. In these systems, the voltage can vary at different points and voltage distribution analysis is important for understanding and controlling the voltage levels in the system.
- Flashover voltage, on the other hand, is a term generally used in insulation systems and refers to a discharge or conductivity phenomenon that occurs between insulation when high voltage is applied. This occurs when the voltage limit of the insulating material or device is exceeded. Flashover voltage is an important parameter for determining insulation durability and is taken into account during insulation design and testing.

High Voltage Tests on Insulators

1. **Power frequency tests**
2. **Impulse test**

Power Frequency Test

- **Dry and Wet Flashover Tests:** In these tests the AC voltage of power frequency is applied across the insulators and increased at a uniform rate of about 2 per cent per second of 75% of the estimated test voltage, to such a value that a breakdown occurs along the surface of the insulator. If the test is conducted under normal conditions without any rain or precipitation, it is called "**dry flashover test**". If the test is done under conditions of rain, it is called "**wet flashover test**". In general, wet tests are not intended to reproduce the actual operating conditions, but only to provide a criterion based on experience that a satisfactory service operation will be obtained. The test object is subjected to a spray of water of given conductivity by means of nozzles. The spray is arranged such that the water drops fall approximately at an inclination of 45° to the vertical. The test object is sprayed for at least one minute before the voltage application, and the spray is continued during the voltage application. The characteristics of the spray are on the right figure.

(i)	Precipitation rate	: 3 ± 10% (mm/min)
(ii)	direction	: 45° to the vertical
(iii)	conductivity of water	: 100 micro Siemens ± 10%
(iv)	water temperature	: ambient ± 15°C

Power Frequency Test

- The International Electrotechnical Commission (IEC) has revised the test procedure and precipitation conditions as follows.
- Average precipitation rate:

vertical component = 1 to 1.5 mm/min

horizontal component = 1 to 1.5 mm/min

limits for individual measurements = 0.5 to 2.0 mm/min

temperature of collected water = ambient temperature ± 15°C

and the conductivity of water corrected to
20°C = 100 ± 15 micro Siemens.

Specifications are being modified for application of 15 positive and 15 negative impulses. Two in each set are allowed to flashover. If more than two flashovers occur in each set, then the insulator is deemed to have failed the test.

Impulse Test

- **Impulse Flashover Test:** Usually, the probability of failure is determined for 40% and 60% failure values or 20% and 80% failure values, since it is difficult to adjust the test voltage for the exact 50% flashover values. The average value of the upper and the lower limits is taken. The insulator surface should not be damaged by these tests, but slight marking on its surface or chipping off is allowed.

Formulas and Calculations Used for Determination of Voltage Distribution

1. Equal Voltage Distribution:

- In this approach, the voltage is evenly divided across each insulator in the string.
- Formula: $V_n = V_t/n$
- Where V_n is the voltage across each insulator, V_t is the total voltage applied to the string, and n is the number of insulators in the string

Formulas and Calculations Used for Determination of Voltage Distribution

2. Capacitance Voltage Divider:

- This method considers the capacitance values of the insulators to calculate the voltage distribution.
- Formula: $V_n = (C_n \ / \ C_t) * V_t$
- Where V_n is the voltage across the nth insulator, C_n is the capacitance of the nth insulator, C_t is the total capacitance of the string, and V_t is the total applied voltage.

Formulas and Calculations Used for Determination of Voltage Distribution

3. Resistance Voltage Divider:

- If the insulators have resistive components, the voltage distribution can be determined using the resistance values.
- Formula: $V_n = (R_n \ / \ R_t) * V_t$
- Where V_n is the voltage across the nth insulator, R_n is the resistance of the nth insulator, R_t is the total resistance of the string, and V_t is the total applied voltage.

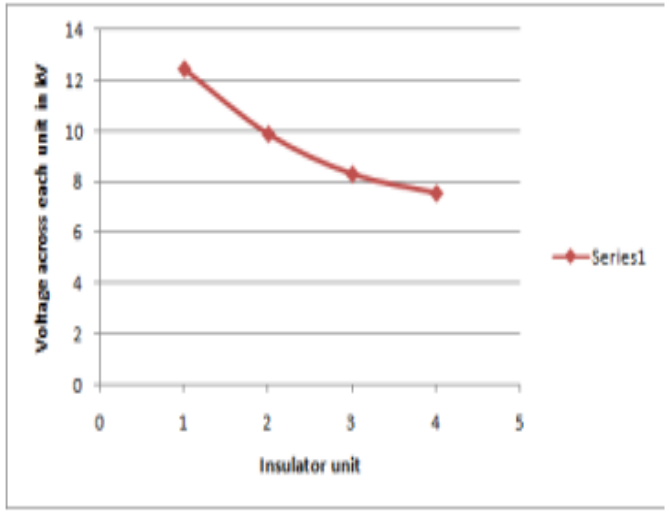
Formulas and Calculations Used for Determination of Flashover Voltage

- **IEC 60060-1** and **IEC 60815** standards provide guidance on flashover voltage estimation based on insulator geometry, pollution level, and material properties.
- **IEEE Std 4** and **IEEE Std 1313** offer recommendations for flashover voltage calculations and testing procedures.
- Laboratory tests, field observations, and experience-based models are also utilized to assess the flashover voltage based on specific insulator string configurations and operating conditions.

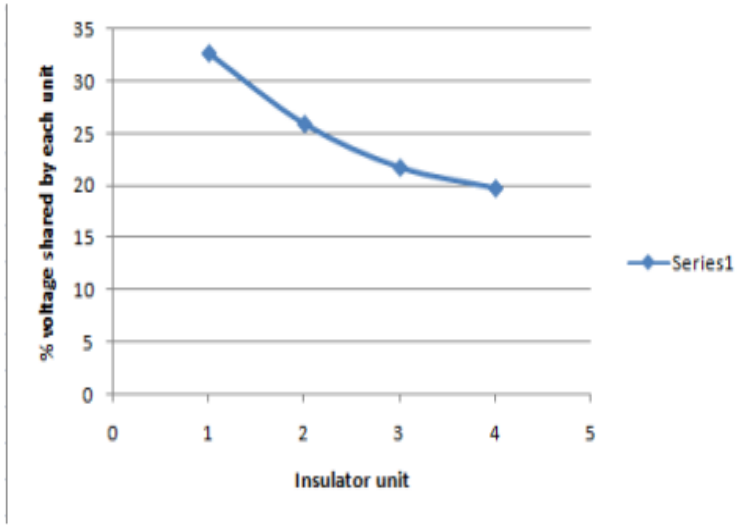
Theoretical Calculation Without Guard Ring

Insulator Number	%age of Voltage Shared	Voltage in kv
1(Line End)	32.6	12.42
2	25.88	9.86
3	21.73	8.28
4(Ground End)	19.76	7.53

Theoretical Calculation Without Guard Ring



Graph between voltage across each unit and given insulator value



Graph between percentage voltage shared by each unit and given insulator value

Theoretical Calculation Without Guard Ring

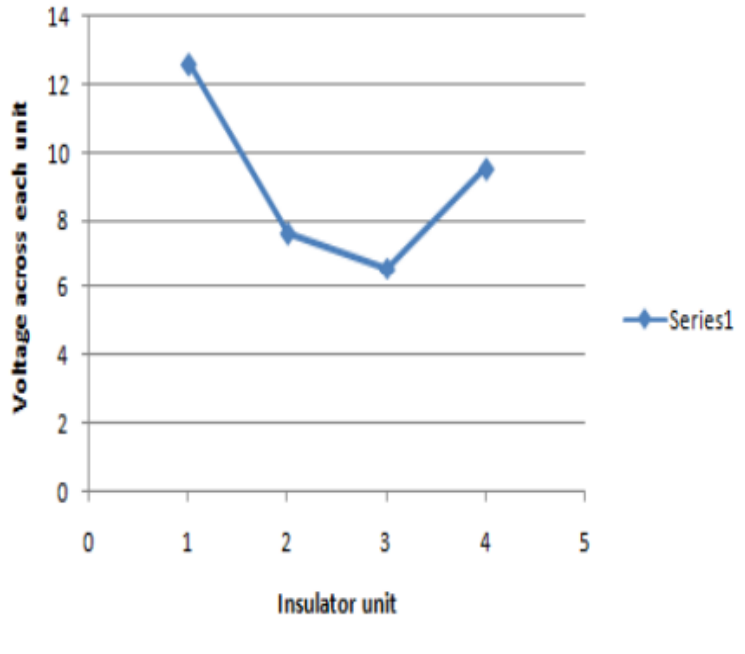
String Efficiency (%)	Total %age of Voltage Shared
76.67	99.8

Standard Method by sphere gap without Using Guard Ring For 2mm Gap Spacing

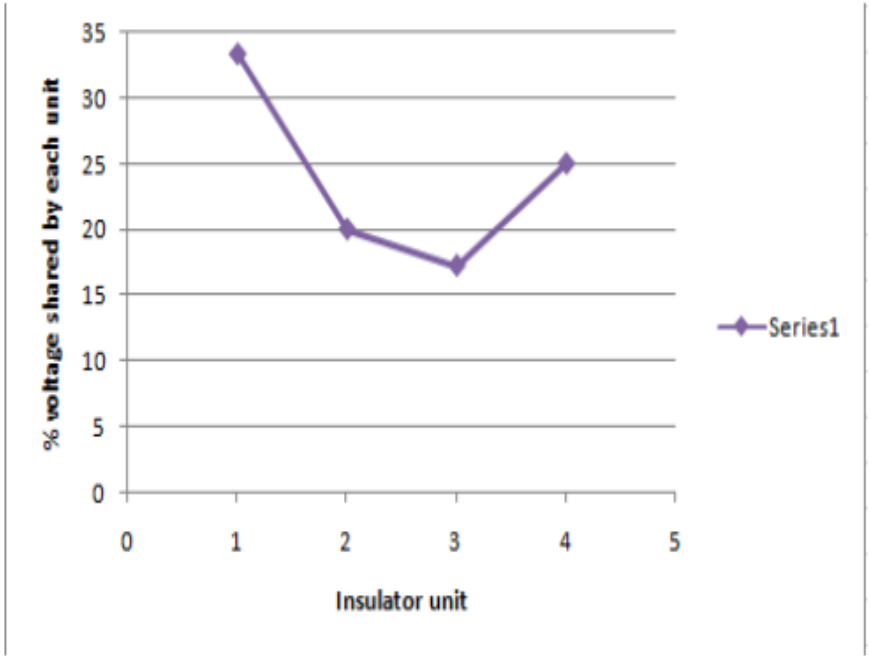
Insulator Number	Breakdown Voltage in kV	%age of Voltage Shared	Voltage in kv
1(Line End)	15	33.33	12.6
2	25	20	7.62
3	29	17.64	6.56
4(Ground End)	20	25	9.52

Theoretical Calculation Without Guard Ring

- String efficiency and total percentage voltage shared by each insulator



Graph between voltage across each unit and given insulator value



Graph between percentage voltage shared by each unit and given insulator value

Theoretical Calculation Without Guard Ring

Standard Method by Using Guard Ring for 2mm Gap Spacing

Insulator Number	Breakdown Voltage in kV	%age of Voltage Shared	Voltage in kv
1(Line End)	15	25.12	13.049
2	16	24.13	13.049
3	15	25.13	13.057
4(Ground End)	15	24.13	13.049

- Voltage distribution across a string of 4 disc insulators calculated practically for 2mm gspacingusingguard rings Thus the voltage is distributed equally by means of guard ring and the different comparison of string efficiency and total percentage of voltage shared for a string of 4 disc insulators is shown below

String Efficiency (%)	Total %age of Voltage Shared
72.06	95.6

Theoretical Calculation Without Guard Ring

- Comparison of String Efficiencyfor a String of 4 disc Insulators

Theoretical	Standard method without guard ring
76.6	72.6

Comparison of Total Percentage of Voltage Shared For a String of 4 disc Insulators

Theoretical	Standard method without guard ring
99.8	95.6

National and Regional Electrical Safety Standards

- High voltage laboratories must comply with the electrical safety standards and regulations applicable in their respective countries or regions. In Turkey, the electrical safety standards and regulations are primarily set by the Ministry of Energy and Natural Resources, the Ministry of Trade, and the Turkish Standards Institution (TSE). These standards and regulations ensure the safe operation, installation, and maintenance of electrical systems and equipment, including those used in high voltage laboratories.
- Turkish Electrical Installation Regulation (TS 845): This regulation outlines the requirements for electrical installations in Turkey, including safety measures, equipment selection, and installation procedures. It covers aspects such as grounding, protection against electric shock, and fire safety.

- Turkish Low Voltage Directive (LVD): The LVD is based on the European Union's Low Voltage Directive and applies to electrical equipment with a voltage range of up to 1,000 volts AC and 1,500 volts DC. It sets requirements for product safety, conformity assessment procedures, and the CE marking for electrical equipment placed on the Turkish market.

National and Regional Electrical Safety Standards

- Turkish Electromagnetic Compatibility Regulation (EMC): The EMC regulation focuses on ensuring that electrical and electronic equipment does not cause interference and operates properly in electromagnetic environments. It covers both emissions and immunity requirements for electrical equipment.

- Turkish Occupational Health and Safety Legislation: Occupational health and safety regulations in Turkey, such as the Occupational Health and Safety Act No. 6331, establish general safety requirements for workplaces, including electrical safety. These regulations cover aspects like risk assessments, electrical equipment maintenance, and personal protective equipment for electrical work.

ITU Fuat Külünk High Voltage Laboratory

The Description and The History of the Lab

- The Istanbul Technical University (ITU) High Voltage Laboratory is a prominent facility located in Istanbul, Turkey. It is an academic laboratory affiliated with Istanbul Technical University, one of the leading technical universities in Turkey. The Laboratory is one of the largest university-based high voltage laboratories in the world and the largest in Turkey. Also according to the voltage level, it is greater than all the high-voltage laboratories in the industry, located in Turkey, and is one of the leading laboratories in the world.

- The name of this laboratory is dedicated to Prof.Fuat Külünk, who was the founding professor of the department and took the first steps to establish this laboratory. In the establishment of the laboratory, Prof. İzzet Gönenç made great efforts. This laboratory, established on an area of 4000 m2, consists of three blocks called A, B, and C:

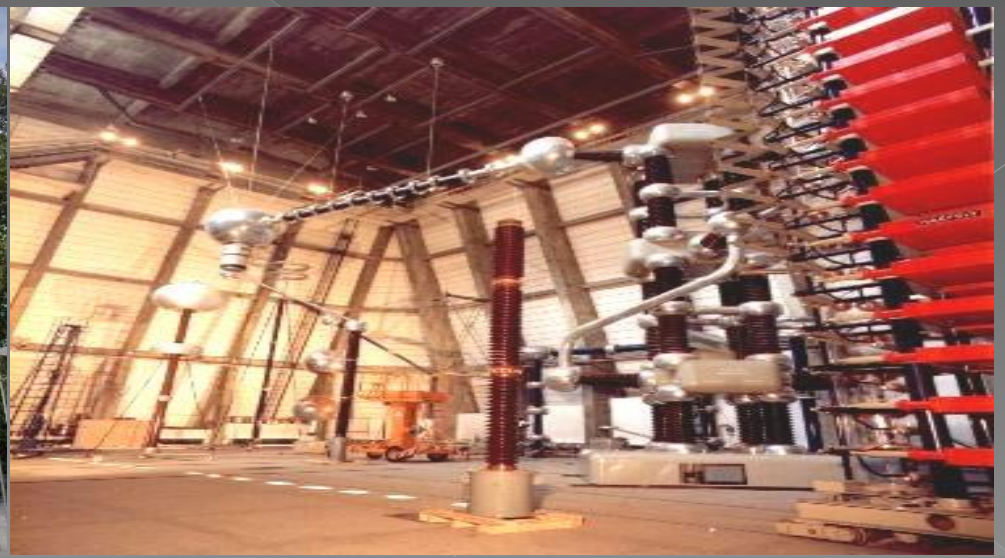
The Description and The History of the Lab

- In block A, there are medium voltage test laboratories and offices, and in block B, equipment capable of performing high voltage tests for 420 kV facilities. In block C, there are power supply systems and panels of the high voltage test equipment.
- Block A is a 3-story building covering an area of 42 x 15 m2. This block, whose construction started in 1961 and was completed in 1963, came into operation in 1964.

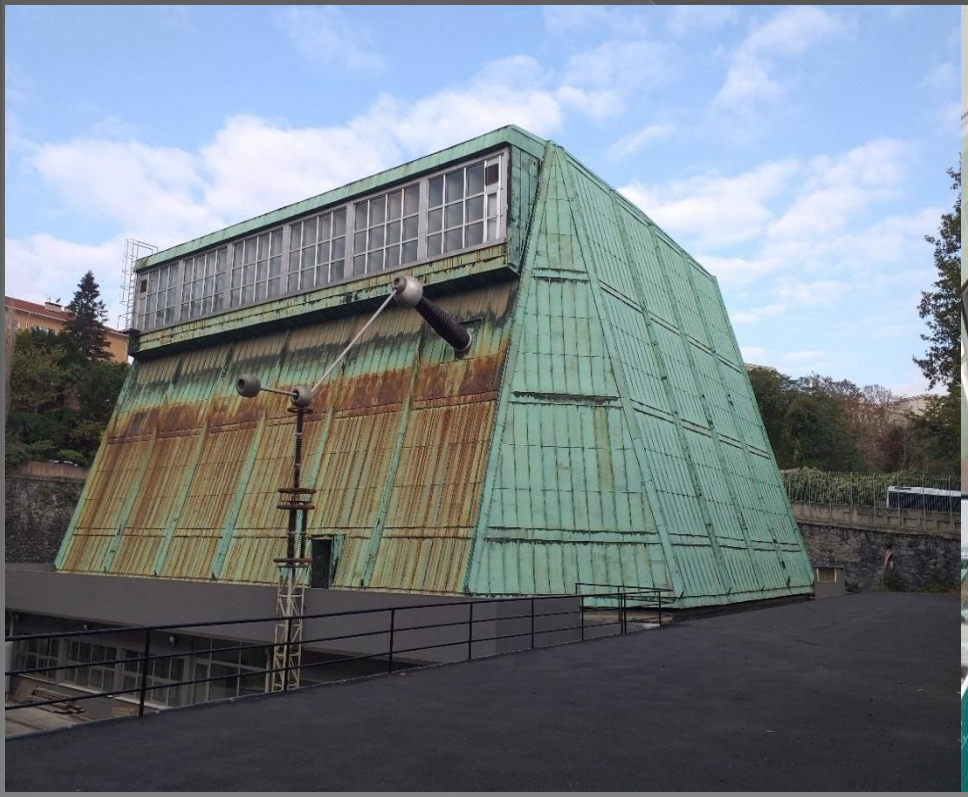
The Description and The History of the Lab

- Block B (Copper Building)
- It is a large laboratory whose dimensions are 26 m x 35 m x 20 m (height), and an area of approximately 1200 m2 with control and study rooms. The laboratory has a 4 m wide and 6 m high external door that allows test objects to enter the laboratory. The laboratory has copperplate shielding and very low grounding resistance. The laboratory is equipped with devices capable of performing high voltage tests for devices with a rate of up to 420 kV.











What are the Main Testing Equipment and Their Capabilities in the Laboratory?

Block A

- In this block, there is an Impulse Voltage Test Laboratory(1000 kV), an AC Test Laboratory(200 kV), a Shielded Test Room(100 kV) and Block A Basement Floor Laboratory.

1000kV Impulse Voltage Test Laboratory

- This laboratory has a 2 x 3.5 m door opening to the C block terrace and a 1.5-ton chain block system that can be disassembled and installed on the terrace of the C block. The laboratory floor was built to withstand 4 tons/m2, and the terrace floor, 7 tons/m2. With this feature, it is easier to transport medium voltage devices to the laboratory.



1000kV Impulse Voltage Test Laboratory

Equipments:

- - Impulse Voltage Generator (1000 kV, 10 kJ, 6 stages),
- - Control desk,
- - Oscilloscope for measuring impulse voltage,
- - Impulse peak voltmeter,
- - Sphere gap (Sphere diameter: 50 cm).

200kV AC Test Laboratory

Equipment:

- - HV Test Transformer (200 kV, 50 Hz, 50 kVA)
- - Capacitive voltage divider (300 kV),
- - Electrostatic voltmeter (100 kV),
- - Sphere gap (Diameter: 50 cm),
- - HV Schering bridge device,
- - Standard high voltage capacitor for tan delta measurement (100 kV, 100,47 pF, gas-insulated)
- - Tower model



100kV Shielded Test Room

- This laboratory is designed for partial discharge and corona experiments. For this reason, the floor, ceiling, and walls of the room are covered with 0.5 mm thick copper plates and screened against electromagnetic fields, and the electricity supply of the room passes through a filter.



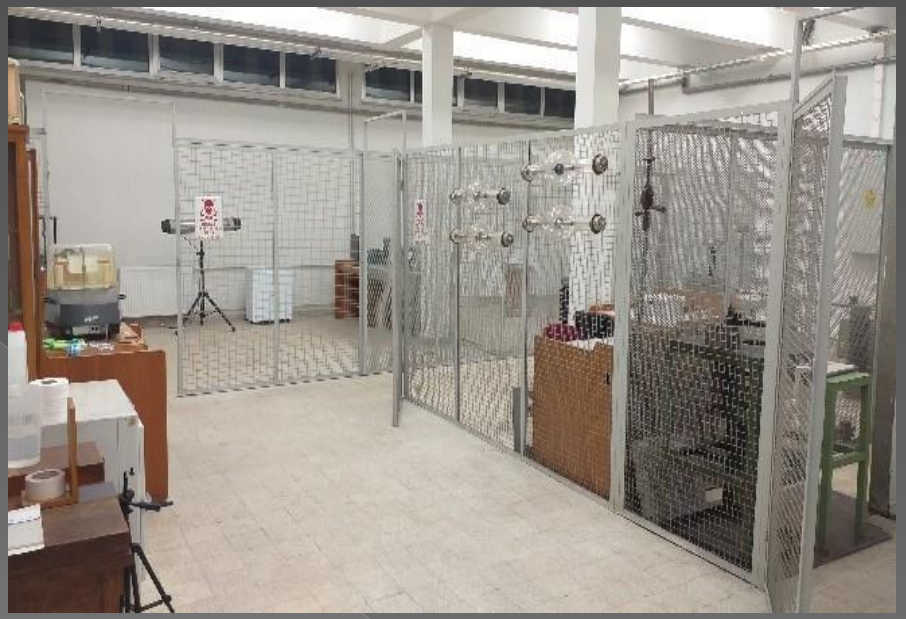
100kV Shielded Test Room

Equipment:

- - 100 kV, 50 Hz, 20 kVA single-phase PD-free test transformer (PD level at 100 kV < 1 mV)
- - Capacitive voltage divider (100 kV),
- - Coupling capacitor (100 kV, 2.5 nF),
- - Advanced partial discharge (PD) measurement and analysis system
- - 90 kV oil dielectric test system,
- - Viscosity meter to investigate the physical properties of oils,
- - Oil aging device,
- - 150oC vacuum oven,
- - 10-4 mmHg vacuum pump.

Block A Basement Floor Laboratory

- A room in the basement of Block A is designed for student experiments and scientific researches. There are Messwandler Bau brand high voltage test sets that allow the generation and measurement of high AC, DC, and impulse voltages.



Block A Basement Floor Laboratory

Equipment:

- - AC generator (transformer) (100 kV, 5 kVA),
- - DC generator (140 kV, 20 mA (single-stage) or 280 kV, 15 mA (double-stage)),
- - Impulse voltage generator (140 kV, 90 Ws (single-stage) or 280 kV, 180 Ws (double-stage)).
- - Portable sphere gaps (Diameters: 10 cm and 25 cm)

1200kV, 50Hz, 1000kVA Cascade High Voltage Test Transformer

- 1.2 MV, 1000 kVA, 3 units;
- Other equipment:
- - 4 kOhm high voltage pre-resistor,
- - Capacitive voltage divider (100 pF),
- - Standard high voltage capacitor for tan delta measurement (800 kV, 68 pF, gas-insulated),
- - Coupling capacitor (2.5 pF, 750 kV)



3600kV, 180kWs 18-stages Impulse Voltage Generator

- Other equipment
- - Capacitive- resistive voltage divider (3600 kV, 4-stages),
- - Resistive voltage divider,
- - 150 MHz and 300 MHz dual-channel oscilloscopes
- - Sphere gap (Dimension: 150 cm)

Block C

- This block contains power supplies for high voltage AC, DC, and impulse generators in Block B.
- - 10 kV supply voltage,
- - 10 / 0.5 kV, 800 kVA transformer,
- - 500 kW, 1500 rpm DC motor,
- - 3-phase 950 kVA synchronous generator for 50-60 Hz,
- - Single-phase 250 kVA synchronous generator for 150 kHz

ITU- Maslak High Voltage Laboratory

- This laboratory, founded in 1995, is located in the ITU Electrical and Electronics Faculty building. The laboratory consists of two separate test rooms.



ITU- Maslak High Voltage Laboratory

- In the laboratory, there are Messwandler Bau brand high voltage test sets that allow the generation and measurement of high alternative current, direct current, and impulse voltages.



ITU- Maslak High Voltage Laboratory

- - High voltage AC generator
- - 100 kV, 5 kVA (single-stage)
- - 200 kV, 5 kVA (double-stage)
- - High voltage DC generator
- - 140 kV, 20 mA (single-stage)
- - 280 kV, 15 mA (double-stage)
- - High voltage impulse voltage genearator
- - 140 kV, 90 Ws (single-stage)
- - 280 kV, 190 Ws (double-stage)

The other equipment:
- Sphere gap (Diameter: 10 cm),
- Sphere gap (Diameter: 25 cm),
- 30 kV electrostatic voltmeter,
- 100 kV electrostatic voltmeter,
- Digital 300 kV impulse peak voltmeter,
- 100 MHz, dual-channel digital oscilloscope,
- 90 kV oil dielectric test system,
- 100 kV oil dielectric test system,
- 150oC vacuum oven,
- 10-3 mmHg vacuum pump,
- Schering bridge test device (1.10-4 < tan delta ± %2 < 10),
- 120 kV, 37 pF SF6-insulated standard capacitor,
- Electric tank test system for analyzing static electric field,
- Electromagnetic Field Meter,
- Various electrode systems.

The Tests Conducted in the Fuat Külünk High Voltage Laboratory

1. **Breakdown Voltage Testing:**
   - This involves determining the voltage at which electrical insulation or insulation materials fail and break down, leading to the formation of an electric arc or current flow.
2. **Dielectric Strength Testing:**
   - It assesses the ability of insulating materials to withstand high electric fields without breaking down. Dielectric strength tests help determine the maximum voltage that an insulator can withstand before failure.

The Tests Conducted in the Fuat Külünk High Voltage Laboratory

3. Partial Discharge Measurement:

- Partial discharges are small electrical discharges that occur within or on the surface of insulation materials. The laboratory may perform tests to detect and measure partial discharges, which can indicate potential insulation problems.

4. Impulse Voltage Testing:

- Impulse voltage tests assess the ability of electrical equipment and insulation systems to withstand sudden voltage surges or impulses, such as lightning strikes or switching operations.

The Tests Conducted in the Fuat Külünk High Voltage Laboratory

5. Power Frequency Withstand Voltage Testing:

- This test evaluates the ability of electrical insulation to withstand continuous voltage stresses at power frequency, ensuring the insulation's integrity during normal operating conditions.

6. Insulation Resistance Testing:

- It measures the resistance of insulating materials or systems to the flow of electrical current, helping evaluate the effectiveness of insulation in preventing leakage currents.

Softwares used in the lab

- Simulation Software: Simulation tools like EMTP-RV (Electromagnetic Transients Program) and MATLAB/Simulink are often used for modeling and simulating electrical power systems. Engineers use these tools to analyze system behavior, simulate fault conditions, and assess the performance of protective devices.
- Measurement and Data Acquisition Software: High voltage laboratories employ measurement and data acquisition software such as LabVIEW and SCADA (Supervisory Control and Data Acquisition) systems. These tools enable real-time data acquisition, visualization, and analysis during high voltage tests.
- Data Analysis and Visualization Tools: Engineers use various data analysis and visualization tools like MATLAB, Python, or Excel for post-processing and analyzing test data. These tools facilitate statistical calculations, data manipulation, and generation of visual representations of the acquired data.

Thank you for listening!

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